Predicting Heart Disease Using SMOTE-Based Data Balancing and Ensemble Learning

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Abstract- **Timely detection of heart disease is a crucial factor in making effective clinical decisions, especially in disadvantaged regions due to the scarcity of healthcare facilities. This research aims to investigate the performance of some supervised machine learning techniques in predicting heart diseases. In this study, regression-based classification, tree-structured machine learning approaches, probabilistic approaches, and approaches using margin values were investigated. Furthermore, the proposed study evaluated some machine learning techniques derived from random bagging and boosting. In addition, class imbalance handling techniques based on the Synthetic Minority Oversampling Technique (SMOTE) were integrated as a solution to the imbalance problem identified in the dataset. Above all, some ensemble machine learning approaches using probability averaging and integration were developed to improve the machine learning performance. Experimental results proved that machine learning approaches based on the random ensemble performed better. In other words, some machine learning approaches based on stacking achieved better performance by reaching a maximum performance with 93.44% accuracy, 0.93 for F1-measure, and 0.90 for precision.**

Keywords- heart disease classification, ROC-AUC, ensemble learning, SMOTE, soft voting, stacking ensemble

I. Introduction

Cardiovascular disease is currently a major global public health issues and it remains to be a significant cause of death annually [1]. The World Health Organization reports that 19.8 million individuals die annually due to cardiovascular disease (CVD) in 2022, accounting approximately 32% of global deaths [1]. This mortality rate is especially high in low and middle-income countries such as Bangladesh, India, and Nepal where sophisticated medical resources are limited and early detection is not easily accessible [2]. As obesity, high blood pressure, smoking, unhealthy eating habits and etc. are increasing the risk of heart disease, it has become extremely important to identify and take preventive measures for those risk at an early stage [1].

Although conventional cardiac diagnostic procedures provide reliable results, their high cost and dependence on specialized clinical resources limit their widespread and rapid use, particularly in resource-constrained settings. In this context, machine learning (ML) has emerged as a promising technology capable of understanding complex, non-linear connections from large clinical datasets to generate accurate predictions. In the past, they employed multiple ML algorithms including Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Gradient Boosting (GB), and XGBoost to anticipate heart disease [3], [4]. No single model is equally effective for all dataset types, even though GB improves predictive accuracy through robust error correction and SVM is useful for high-dimensional data analysis. For instance, SVM frequently needs more processing power, Decision Trees are prone to overfitting, and Logistic Regression is unable to identify non-linear relationships. Furthermore, a number of prior research investigations have identified deficiencies in the model's interpretability, class-inequality neglect, inadequate data preprocessing, and a lack of hyper parameter optimization, all of which diminish clinical applicability.

The study used RandomizedSearchCV for hyperparameter tuning, class balancing, and continuous data preprocessing in order to lessen these constraints. LR, DT, RF, KNN, SVM, and GB were among the models whose performance was examined; as an ensemble model, SVM and GB & RF produced the best results. The suggested SVM, GB ensemble model combines the advantages of both algorithms: GB improves the model's robustness through consistent error correction, while SVM establishes clear decision boundaries in the higher dimensional feature space. A more accurate and consistent prediction is offered by this ensemble approach [5]. However, RF facilitates robust feature interaction learning and variance reduction.

The purpose of this paper is to create a reliable support system for researchers and medical practitioners engaged in the diagnosis and treatment of heart disease. As can be seen, the findings of this study illustrate the effective applicability of ensemble learning algorithms and models and also emphasize the importance of hyperparameter tuning and efficacious preprocessing techniques for the precise prediction of heart disease.

II. Literature Review

Traditional statistical models were a major component of early CVD prediction research. The most noteworthy study was by Detrano et al. [6], who used logistic regression to

present a probabilistic model for predicting coronary artery disease with a 77% accuracy rate. This model subsequently served as a standard for other research. Nevertheless, logistic regression was limited by its incapacity to capture the intricate relationships among patient age, cholesterol, and other variables. Subsequently, scientists used non-linear machine learning techniques.

Showman et al. [7] achieved about 87% accuracy using KNN algorithm. This demonstrates that non-parametric models learn nonlinear relationships in data finely. Ensemble learning becames very popular during the period 2017-2020. This era was characterized by comprehensive reviews, such as that by Al-Janabi et al. [12], explaining the effectiveness of advanced techniques likes SVM and ANN. Latha & Jeeva [8] proved that bagging and boosting always give better results than single classifiers. Another important work of this time is the Hybrid LR-NN model of Amin et al. [9], which gives 89.40% accuracy.

Another notable contribution was by Mohan et al. [11], who introduced a sophisticated Hybrid Random Forest and Linear Model. However, most of the studies ignore the class imbalance problem, where the models become biased due to the low number of positive cases. A major shift was noted in the classification by the researchers between the years 2019 and 2024, as they employed the use of big data frameworks and features of ECG signal classification in improving classification efficiency. Allersan and Younes [13] extracted heartbeat features in detail using a large dataset of 205,146 records obtained from an ECG signal classification model and implemented a random forest classifier using the Apache Spark programming environment. The data produced an impressive accuracy value of 97.98%, confirming the strength of large-scale and high-resolution biomedical data in detecting heart disease.

The latest literature suggests that further improvements in terms of accuracy in data models should be achieved by optimized hybridizations. This includes automated selection models, enhancing data features, and using deep learning to identify complex data interactions, among other things. Focusing on optimization: Studies by Kadhim and Radhi [14], Solanki et al. [15], and Jevin et al. [16] point to the use of highly refined forms of machine learning. Hyperparameter tuning is a common mechanism in optimization, especially in SVM and random forest algorithms. Hybrid Deep Learning Complexity: A good example of how complex data models have become in recent times can be gleaned from Al-Alsheikh et al. [17]. Each of the key data and modeling constraints faces pressure with their hybrid, multi-stage pipeline modeling, known as ML-HDPM. Feature Engineering: Notably, the most relevant features are identified automatically with the integration of a combination of genetic algorithms and iterative feature elimination in data processing, thereby ensuring robustness in identifying issues by removing noise within the data. Data Balancing: The main problem of class balancing was addressed by utilizing a specialized under sampling clustering oversampling technique, known as USCOM. Classification: Ultimately, classification is achieved by using multilayer deep convolutional neural network.

However, despite notable progress in deep learning and large data models, a key challenge persists in obtaining high prediction accuracy with this UCl heart disease dataset while ensuring the model is simple, interpretable, and appropriate for deployment in hospitals. The proposed study is focused on meeting this crucial objective:

- To mitigate class imbalance in clinical heart disease data, SMOTE is employed as a crucial preprocessing technique to generate synthetic minority samples, reduce prediction bias, and substantially enhance recall, sensitivity, and F1-score, an aspect underexplored in existing studies.
- The proposed framework employs an optimized soft voting ensemble that aggregates probability outputs from multiple robust classifiers, ensuring stable and reliable predictions, reducing individual model limitations, achieving high accuracy with low computational cost, and outperforming conventional voting and feature-selection-based approaches.
- Unlike deep CNN models or large big-data frameworks, the proposed method is- Efficient on standard, low-cost computers. Dependent only on commonly available clinical attributes. Practical and deployable in real healthcare environments.

Here, Table - I represent the accuracy and the best model of the related works as we described above.

TABLE I. SUMMARY OF RELATED WORKS

Paper Authors	Publish Year	Best Model	Accuracy
Detrano et al. [6]	1989	Logistic Reg	77.00%
Showman et al. [7]	2012	KNN	87.00%
Amin et al. [9]	2019	Hybrid LR-NN	89.40%
Alarsan & Younes [13]	2019	Random Forest	97.98%
Spencer et al. [10]	2020	Random Forest	89.70%

III. METHODOLOGY

In this research, it introduces a Machine Learning framework for classifying heart disease, encompassing the stages of gathering datasets, preprocessing, transforming and scaling features, training models, and evaluating performance. After the preprocessing is completed, multiple supervised models including LR, DT, RF, GB, SVM, KNN, AdaBoost, and Naive Bayes (NB), and soft voting and stacking ensemble methods are applied to enhance performance.

A. Dataset Description

This dataset was obtained from the UCI Machine Learning Repository and contains 303 patient records featuring 14 attributes, which consist of 13 independent variables and one target variable. The characteristics reflect key medical and public health elements associated with heart disease, including age, gender, chest pain, blood pressure, cholesterol levels, electrocardiographic results, heart rate, exercise-related indicators, and thalassemia condition.

B. Data Preparation and Preprocessing

One of the most important stages of the machine learning process is data preparation. The following preprocessing steps are carried out in a sequential manner while closely following the defined classification code-

At first, NaN was used to represent missing information in the dataset instead of invalid values. Since the median is less sensitive to outliers when compared to the mean, imputation of missing values was done using the median. Numerical columns containing string values were converted to proper numeric format. To avoid biased learning and data leakage, duplicate patient records were examined and eliminated. Following the transformation of the original target column (num), which included various disease levels, into a binary target- 0 indicating no heart disease and 1 indicating its presence, the original num column was eliminated. The IQR method was used to cap extreme values, and boxplots were used to find outliers in numerical features. Boxplots after treatment verified less impact from outliers.

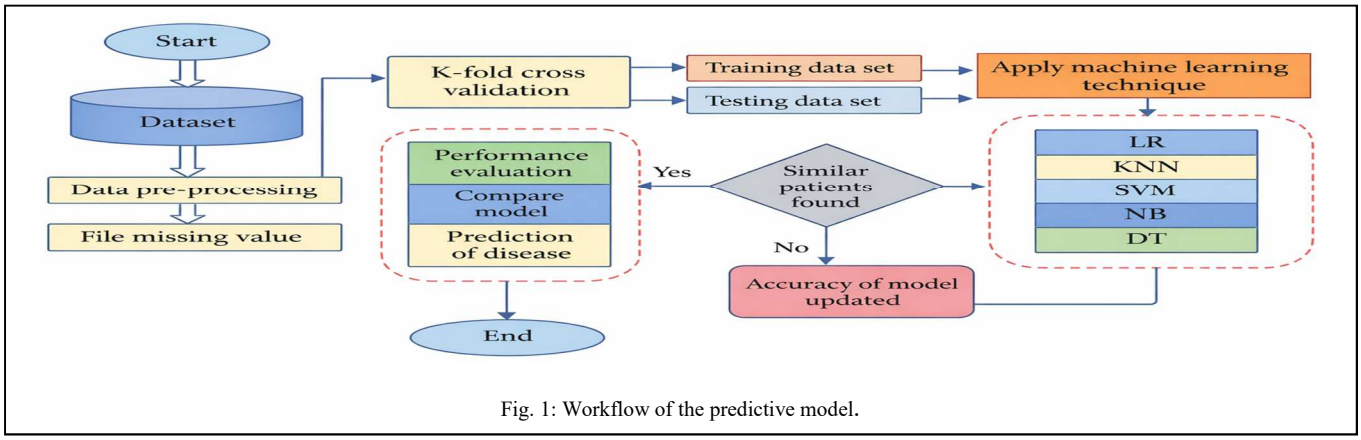


Fig. 1: Workflow of the predictive model.

One-hot encoding helps to convert categorical features of the dataset into numerical format. In this process, each categorical variable is transformed into multiple binary features, which enables machine learning algorithms handle categorical variables without any ordinal bias. Feature Scaling and Normalization is used to ensure equal contribution from all features, Min-Max Scaling is applied to the numerical features, which rescales the data into the range [0,1]. The Min–Max scaling formula:

$$X' = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

Following the preprocessing stage, the dataset was separated into a target vector (Y) with binary class labels and a feature matrix (X) with independent variables. Data was then split into train and test sets to assess model generalization. To correct class imbalance, the SMOTE technique, which creates synthetic minority class samples via nearest-neighbor interpolation rather than by replicating any already existing data, was used. This resulted in a balanced dataset for better model performance. For a given minority class sample X_i , let the synthetic data point be represented as X_{new} , which is generated using the following equation:

$$x_{new} = X_i + \lambda * (x_{nn} - x_i) \quad (2)$$

The Pearson Correlation Coefficient is used to generate a correlation heatmap that examines the relationship between target variables and numerical features. The linear relationship between the features is visually represented in this analysis.

C. Classification Process

Having completed this data preprocessing and transformation step, the generated data was then utilized to perform a supervised classification of each entry in the patient data set to fall into one of two classes: presence or absence of heart disease. The feature matrix can be expressed as:

$$X = \{x^1, x^2, x^3 \dots \dots, x^n\}$$

and the relevant target labels as:

$$Y = \{y^1, y^2, y^3, \dots \dots \dots . y^n\} \quad y^i \in \{0,1\}$$

Each machine learning classifier learns a mapping function:

$$f: X \to Y$$

D. Machine Learning Models

Several different algorithms based on supervised learning have been applied. We will describe what each of the mathematical models does:

LR: It is a statistical classification method; this technique is generally used to solve binary classification problems. In this model, the probability of a positive class will be calculated by the logistic function. The goal of the hypothesis will be as follows-

$$p\ (y = 1 \mid x) = \frac{1}{(1 + e^{-z})} \quad (3)$$

where,

$$z = \beta_0 + \Sigma\ (\beta_i\ x_i),\ i = 1\ to\ n$$

The parameters of the model are learned by minimizing the binary cross-entropy loss function, it can be expressed by the following equation-

$$X = -\frac{1}{N} \Sigma\ [\ y_i\ log(\hat{y}_i) + (1 - y_i)\ log\ (1 - \hat{y}_i)] \quad (4)$$

DT: This is a rule-based classifier that uses a recursive partitioning approach on a given dataset based on features.

$$Gini = 1 - \sum_{i=-1}^{c} p_i^2 \quad (5)$$

The Gini Index represents a widely used measure of impurities in a decision tree, where p_i describes the proportion of samples associated with class i. After splitting on a particular node, a feature associated with the least value of Gini Index is selected. This continues until a predefined stopping criterion is met; this may include a predetermined maximum depth of trees created by now or a situation where a leaf node with zero impurities has been created.

RF: This is an ensemble learning method, which means that multiple decision trees are combined into a single model by the application of bootstrap sampling and random feature selection to improve generality. Each tree is learned on a bootstrap sample, and prediction is by majority vote:

$$\hat{y} = mode\ (y^1, y^2, \dots y^T) \quad (6)$$

where T represents the quantity of trees. Random Forest decreases variance and enhances generalization.

GB: GB builds models in a series, where each new model decreases the loss of the prior ensemble. The model's forecast at iteration t is:

$$\hat{y}^{(t)} = \hat{y}^{(t-1)} + \eta\, f_t(x) \quad (7)$$

where, η represents the learning rate and f_t(x) denotes the weak learner. The objective function is:

$$L = \sum_{i=1}^{N} \mathbb{I}(y_I, \hat{y}_i) + \Omega(f) \quad (8)$$

SVM with Hyperparameter Tuning: SVM is a method that seeks to identify an ideal hyperplane that maximizes the gap between different classes. The issue of optimization is expressed as:

$$min \frac{1}{2} |w|^2 + C \sum_{i=1}^{N} \xi i \quad (9)$$

Where, C is the regulation parameter and ξi is slack variables. The kernel functions enable the SVM so that they handle non-linear classification tasks. Hyperparameters tuned: regularization parameter C, kernel type (e.g., linear, polynomial, RBF), and kernel-specific parameters (e.g., γ).

The Soft Voting Ensemble: The Soft Voting Ensemble, an ensemble learning method based on classical ensemble learning theory. Instead of depending only on predicted class labels as in hard voting, this method averages the predicted class probabilities of several base classifiers. The final prediction is computed as:

$$\hat{y} = arg \max_{c} \sum_{m=1}^{M} P_m(y = c \mid x) \quad (11)$$

The Stacking Ensemble: Stacking ensemble, known as stacked Generalization, is an ensemble learning technique that employs a meta-learner to integrate multiple base classifiers. This approach consists of initially training base models independently, then employing their predictions as input features for an upper-level model that learns to optimally combine the outputs of the base classifiers.

Let $f_1, f_2, \dots, f_m$ denote the base classifiers. The meta-model prediction is defined as:

$$\hat{y} = g(f_1(a), f_2(a), \dots, f_m(a)) \quad (12)$$

where, $f_m(a)$ is the prediction of the m-th base classifier and g(.) denotes the meta-learner.

In addition to these popular models, other classifiers were used to evaluate their effectiveness. Hence, these classifiers were used to ensure fair evaluation of their efficiency. K-Nearest Neighbors (KNN) and AdaBoost were used along with their hyperparameters to make these classifiers more efficient. Naive Bayes was used as a probabilistic model due to its simplicity and computational efficiency. The use of these classifiers will guarantee the reliability of this proposed method with regard to consistency under a variety of learning situations.

E. Performance Evaluation Metrics

Model performance was tested by using an error matrix that categorized the outcomes as correctly identified cases and misclassification errors. We computed standardized indices by monitoring accurate positive detection and correct negative identification: overall Accuracy and Precision. We paid particular attention to Sensitivity (Recall) to assess the performance of the system to capture actual cardiac instances. As the clinical data was imbalanced, we used the F1-score as a single metric in case of class imbalance, which is the harmonic average between precision and sensitivity. Finally, AUC was used to measure the ability of models to effectively discriminate between cardiac and non-cardiac patients.

$$Accuracy = \frac{TP+TN}{TP+FP+TN+FN} \quad (13)$$

$$Precision = \frac{TP}{TP+FP} \quad (14)$$

$$Recall = \frac{TP}{TP+FN} \quad (15)$$

$$F1 = 2 * \frac{Precision \times Recall}{Precision \times Recall} \quad (16)$$

$$TPR = \frac{TP+TN}{TP+FN} \quad (17)$$

$$FPR = \frac{FP}{FP+TN} \quad (18)$$

IV. Result and Discussion

A. Performance Comparison and Analysis of Individual Classification Models:

The performance comparison for each individual classifier is re-presented in Table II, using F1-score, accuracy, precision, and recall as evaluation metrics.-

TABLE II. PERFORMANCE COMPARISON OF INDIVIDUAL MACHINE LEARNING MODELS

Model Name	Accuracy	Precision	Recall	F1-score
KNN	**0.9180**	0.9305	0.9180	0.9180
GB	0.9016	0.9096	0.9016	0.9017
RF	0.8852	0.8860	0.8852	0.8854
LR	0.8852	0.8900	0.8852	0.8854
AdaBoost	0.8525	0.8640	0.8525	0.8525
SVM	0.8525	0.8532	0.8525	0.8526
DT	0.7705	0.7948	0.7705	0.7690
NB	0.6230	0.6946	0.6230	0.6014

In the result section, the KNN algorithm is the one which gained prominence with a notable score of 0.9180 owing to its approach of scaling and normalizing the input variables. Next, after Gradient Boosting at 0.9016, followed by a mere difference with Logistic Regression and Random Forest, which scored 0.8852, is SVM and AdaBoost at 0.8525, and then we have the Decision Tree score at 0.7705, a very low score due to overfitting, and lastly, the lowest score was for Naive Bayes at 0.6230, as it was not able to cut through the connections formed between the features and the data.

B. Performance Analysis of Ensemble Models

Ensemble techniques were employed to enhance the predictive performance even further. Table III indicates some of the accuracy metrics obtained for these models –

TABLE III. PERFORMANCE OF ENSEMBLE MODELS

Ensemble Model	Accuracy
Stacking Ensemble (SVM+GB+RF)	**0.9344**
Soft Voting Ensemble	0.9344
Stacking Ensemble (KNN+GB+RF)	0.9180

In the results, Soft Voting Ensemble and the Stacking Ensemble achieved the highest accuracy of 0.9344, which is higher than all the individual classifiers. This improved result shows that combining multiple classifiers reduces the drawbacks of individual models and increases the model's capacity for generalization.

C. Confusion Matrix Analysis of Ensembles Models:

Confusion matrices for the ensemble models have been calculated in order to examine them in detail. Figure 3 represents the confusion matrix for the Stacking Ensemble Model (SVM + GB + RF).

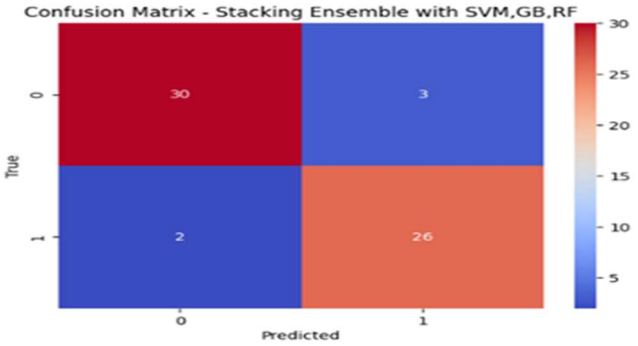


Fig. 3: Stacking Ensemble (SVM + GB + RF)

The above confusion matrix states that the model is actually predicting correctly that there are 30 instances that don't have heart diseases and 27 instances that have heart disease. It's also stating that there are merely 3 false positives and 1 false negative, and this issue holds a lot of importance in terms of medical diagnostics. A different setting of clustering algorithms, like KNN, GB, and RF (Fig-4), was implemented and proved its potential, though increasing false positives slightly.

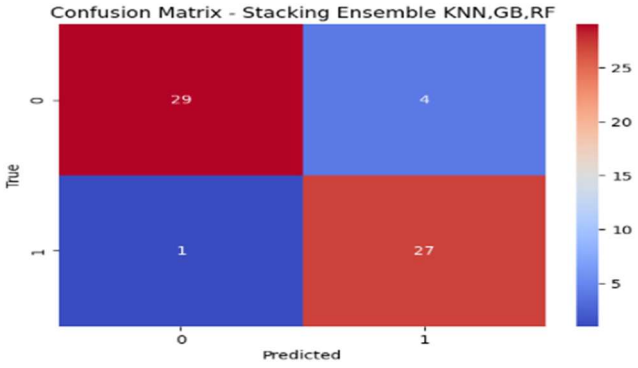


Fig. 4: Stacking Ensemble (KNN + GB + RF)

The performance confusion matrix of the Soft Voting (SVM+GB+LR) has been shown in Fig 5, which showing minimal misclassification and strong identification power for detecting actual heart disease cases.

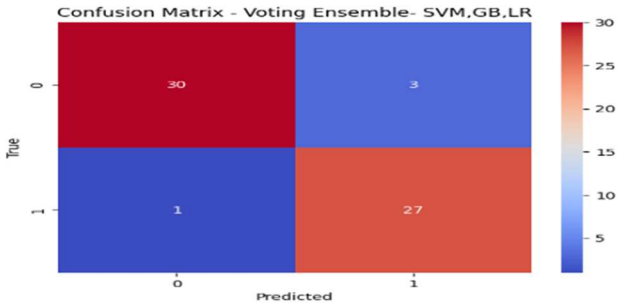


Fig. 5: Soft Voting Ensemble

D. Classification Report Analysis of Ensemble Models:

Stacking Ensemble (SVM + GB + RF): For class 0, the non-heart disease class, the stacking ensemble achieved a precision of 0.97, recall of 0.91, and F1-score of 0.94. On the other hand, for the heart disease category, class 1, it achieved 0.90 in precision, 0.96 in recall, and 0.93 in F1-score. It shows that the overall accuracy of the stacking ensemble is 0.93, where both the macro-average and weighted-average F1-scores also stand at 0.93, showing consistent performance across both classes.

Soft Voting Ensemble: The classification metrics of the stacking ensemble and the soft voting ensemble were exactly the same. Precision, recall, and F1-score are 0.97, 0.91, and 0.94 for class 0 and 0.90, 0.96, and 0.93 for class 1. For the voting ensemble, the macro-average and weighted-average F1-scores are 0.93, showing an overall accuracy of 0.93. This consequently established a fact that both models are equally effective regarding performance and prediction accuracy on every class.

E. Comparative Analysis

The experimental results have proven that using ensemble classifiers performs better in heart disease prediction compared to individual classifiers. Stacking classifiers that use SVM, GB, and RF, along with soft voting classifiers, produced comparable outcomes, achieving an accuracy of 0.93 and a macro/weighted average F1-score of 0.93. However, stacking classifiers should be given more preference due to its hierarchical learning concept involving a meta-learner. The ensemble classifier can be more flexible, robust, and scalable compared to soft voting, where averaging is done to obtain possible outputs from the classifier. Hence, stacking is identified as a potential model.

V. CONCLUSION AND FUTURE WORK

The effectiveness of ensemble-based machine learning models in predicting coronary artery disease (CAD) is demonstrated by this study. After intensive pre-processing and evaluation of eight classifiers with the Ensemble Technique, the Ensemble achieved best results in terms of Accuracy, at 93.44%, along with an F1 score of 0.93, hence providing the desired balance between Accuracy and Performance to justify Ensemble as a Clinical Situation Assistance Technique.

With a view to further improve and advance the model's efficiency and understandability, upcoming studies will concentrate on the extension of the existing data by incorporating multi-site clinical information and exploring new techniques such as deep learning, ensemble-hybrid, and

explainable artificial intelligence. Moreover, to further increase the usability of the proposed approach, it will also be integrated into electronic health records, or EHRs.

VI. References

[1] World Health Organization, "Cardiovascular diseases (CVDs)," *World Health Organization* , Jul. 31, 2025. https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)

[2] C. J. L. Murray *et al.*, "Global Burden of 87 Risk Factors in 204 Countries and territories, 1990–2019: a Systematic Analysis for the Global Burden of Disease Study 2019," *The Lancet*, vol. 396, no. 10258, pp. 1223–1249, Oct. 2020, doi: https://doi.org/10.1016/s0140-6736(20)30752-2.

[3] S. Patidar, D. Kumar, and D. Rukwal, "Comparative Analysis of Machine Learning Algorithms for Heart Disease Prediction," 2022. doi: https://doi.org/10.3233/ATDE220723.

[4] M. Ozcan and S. Peker, "A classification and regression tree algorithm for heart disease modeling and prediction," *Healthcare Analytics*, vol. 3, p. 100130, 2023, doi: https://doi.org/10.1016/j.health.2022.100130.

[5] A. Tiwari, A. Chugh, and A. Sharma, "Ensemble framework for cardiovascular disease prediction," *Computers in Biology and Medicine*, vol. 146, p. 105624, Jul. 2022, doi: https://doi.org/10.1016/j.compbiomed.2022.105624.

[6] R. Detrano *et al.*, "International application of a new probability algorithm for the diagnosis of coronary artery disease," *The American Journal of Cardiology*, vol. 64, no. 5, pp. 304–310, Aug. 1989, doi: https://doi.org/10.1016/0002-9149(89)90524-9.

[7] M. Shouman, T. Turner, and R. Stocker, "Applying k-Nearest Neighbour in Diagnosing Heart Disease Patients," *International Journal of Information and Education Technology*, vol. June 2012, pp. 220–223, 2012, doi: https://doi.org/10.7763/ijiet.2012.v2.114.

[8] C. B. C. Latha and S. C. Jeeva, "Improving the accuracy of prediction of heart disease risk based on ensemble classification techniques," *Informatics in Medicine Unlocked*, vol. 16, p. 100203, 2019, doi: https://doi.org/10.1016/j.imu.2019.100203.

[9] M. S. Amin, Y. K. Chiam, and K. D. Varathan, "Identification of significant features and data mining techniques in predicting heart disease," *Telematics and Informatics*, vol. 36, pp. 82–93, Mar. 2019, doi: https://doi.org/10.1016/j.tele.2018.11.007.

[10] R. Spencer, F. Thabtah, N. Abdelhamid, and M. Thompson, "Exploring feature selection and classification methods for predicting heart disease," *DIGITAL HEALTH*, vol. 6, p. 205520762091477, Jan. 2020, doi: https://doi.org/10.1177/2055207620914777.

[11] S. Mohan, C. Thirumalai, and G. Srivastava, "Effective Heart Disease Prediction Using Hybrid Machine Learning Techniques," *IEEE Access*, vol. 7, no. 1, pp. 81542–81554, 2019, doi: https://doi.org/10.1109/access.2019.2923707.

[12] M. Aljanabi, M. Qutqut, and M. Hijjawi, "Machine Learning Classification Techniques for Heart Disease Prediction: A Review," *International Journal of Engineering and Technology*, vol. 7, pp. 5373–5379, Oct. 2018, doi: https://doi.org/10.14419/ijet.v7i4.28646.

[13] F. I. Alarsan and M. Younes, "Analysis and classification of heart diseases using heartbeat features and machine learning algorithms," *Journal of Big Data*, vol. 6, no. 1, Aug. 2019, doi: https://doi.org/10.1186/s40537-019-0244-x.

[14] M. Abood Kadhim and A. M. Radhi, "Heart disease classification using optimized Machine learning algorithms," *Iraqi Journal for Computer Science and Mathematics*, vol. 4, no. 2, pp. 31–42, Feb. 2023, doi: https://doi.org/10.52866/ijcsm.2023.02.02.004.

[15] A. Solanki, A. Vardhan, A. Jharwal, and P. N. K. I, "Heart Diseases Prediction Using Machine Learning," in *2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, pp. 1–6. doi: https://doi.org/10.1109/ICCCNT56998.2023.10307839.

[16] M. Jevin, H. Jayant, R. Sanjay, V. Hemasai, and P. Venkatasrinivas 5 1-Assistant, "Heart Disease Identification Method Using Machine Learning Classification in E-Healthcare," 2023. Available: https://ijarasem.com/admin/img/148_Heart_NC.pdf

[17] H. A. Al-Alshaikh *et al.*, "Comprehensive evaluation and performance analysis of machine learning in heart disease prediction," *Scientific Reports*, vol. 14, no. 1, p. 7819, Apr. 2024, doi: https://doi.org/10.1038/s41598-024-58489-7.